

★ SPECIAL • LECTURE

The AI-Native Asset Manager

Multi-agent investment firms — a team of AI agents that research, trade, and check risk

The capstone of this course: what if the whole investment process ran as a team of cooperating AI agents? We meet the roles, the workflow, the tech stack — and the hard governance questions.

Why Everyone Is Talking About This

Agents jumped from demos to the investment mainstream

In the last year, AI agents moved from research toys into real investment workflows.

The idea: don't just use AI as a tool — rebuild the firm AROUND a team of cooperating agents.

What's driving it

- LLMs can now read, reason, and use tools (Week 12).
- Agents can act in a loop toward a goal (Weeks 13, 15).
- A flood of investment in agentic startups.

The honest reality check

- Most pilots still fail to reach production.
- Trust in fully autonomous AI actually fell in 2025.
- The winners pair ambition with strong governance.

Where We Are Going Today

From the idea to the team, the build, and the risks

1

What 'AI-native' really means

The shift from AI-as-a-tool to a firm built around agents that run the workflow.

2

The agent investment team

The roles — analyst, researcher, trader, risk, manager — and how a decision flows through them.

3

Building it, and governing it

The tech stack, the promise, and the dangers that make humans and guardrails essential.

This ties the whole course together: LLMs (W12), RL (W13), agents (W15), and our three apps.



1

What 'AI-Native' Means

From tool to teammate

AI-Augmented vs. AI-Native

Is AI a helper you prompt, or the team that runs the work?

Two Ways to Run an Asset Manager

AI-augmented (today)

People do the work; AI is a helper tool.

A chatbot summarizes a report.

A model flags a few trades.

Each tool is used one prompt at a time.

A human prompts every single step.

Useful, but the workflow is still manual.

AI-native (emerging)

A team of AI agents runs the workflow.

Each agent has a job: research, trade, risk.

They hand work to each other automatically.

Humans set goals and approve big moves.

People oversee OUTCOMES, not every step.

Built around agents, not screens and forms.

The leap: from prompting a tool at each step to overseeing a team of agents that run the workflow.

Assistant vs. Agent

The crucial difference is who drives

An AI assistant

- Waits for you to prompt it, every step.
- Answers, then stops.
- You stitch the steps together by hand.
- Like the Week 12 chatbot.

An AI agent

- Given a goal, it plans the steps itself.
- Uses tools, reads results, decides what's next.
- Runs a multi-step workflow with less hand-holding.
- Like the Week 15 agent loop, specialized.

Agents shift people from EXECUTING tasks to OVERSEEING outcomes — the heart of 'AI-native'.

2

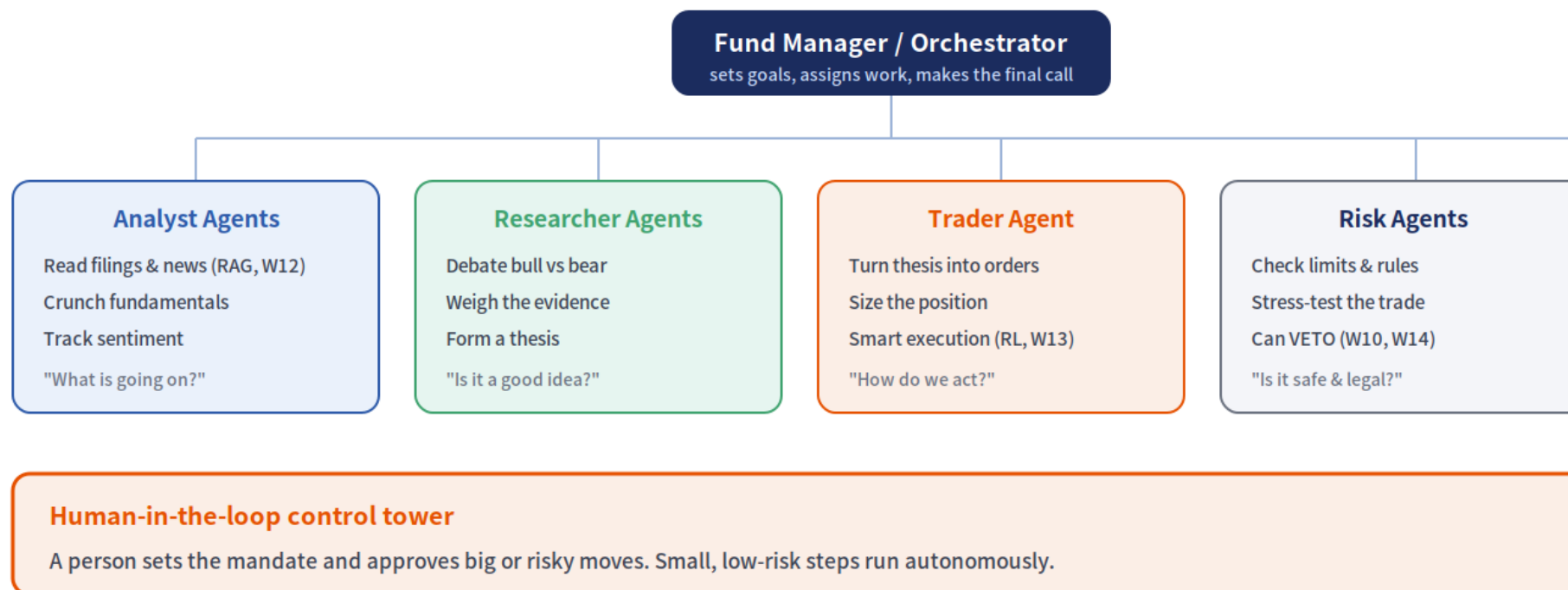
The Agent Investment Team

Roles and the decision flow

The Agent Investment Team

Mirror a real investment firm — one agent per role

The Agent Investment Team — Each Agent a Specialist



This mirrors real research like TradingAgents: analyst, researcher, trader, risk, and a fund-manager agent.

Why a TEAM Beats One Big Agent

Specialists and debate, just like real firms

You could ask one giant model 'should I buy this stock?' — but you'd get one fuzzy opinion.

Splitting the job into specialist roles that hand off and DEBATE produces sharper, checkable decisions.

The benefits

- Each agent focuses on one job and does it well.
- Bull-vs-bear debate surfaces weak arguments.
- Separating 'decide' from 'execute' adds safety.

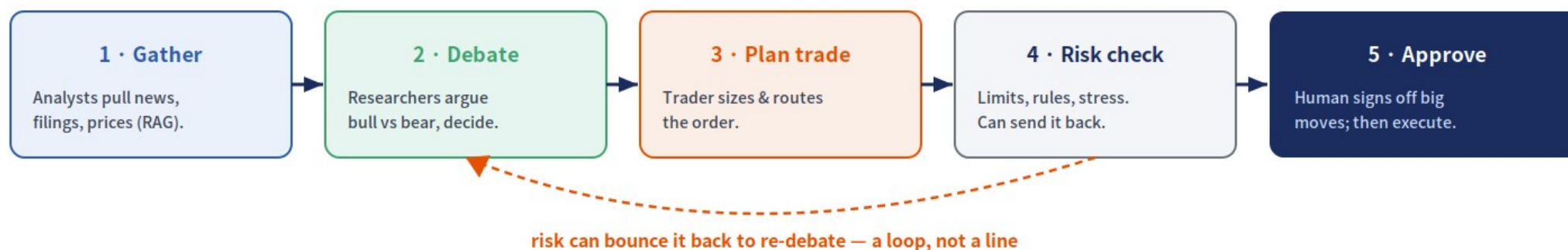
Echoes the course

- Process beats genius (Week 1) — at the firm level.
- A risk agent that can VETO (Weeks 10, 14).
- Each role reuses tools you already built.

How a Decision Flows Through the Firm

Gather, debate, plan, risk-check, approve — with loops

How a Decision Flows Through the Firm



Every step leaves a written trail

Because agents reason in text, you can read WHY each decision was made — built-in explainability (Week 14).

That audit trail is what makes the whole thing trustworthy enough for real money.

Because agents reason in text, every step leaves a readable audit trail — built-in explainability.

The Roles, and the Course Tools They Use

Every week of this course shows up somewhere

1

Analyst & Researcher agents

Read filings and news with RAG (W12); gauge sentiment; debate a thesis. The 'finance chatbot' app, grown up.

2

Trader agent

Turn the thesis into sized orders; route them with smart execution and RL (W13). Builds robust portfolios (W11).

3

Risk agent & Fund-manager / Orchestrator

Risk checks limits, fairness, and overfit backtests (W10, W14) and can veto. The manager assigns work and decides.

Fraud-style anomaly detection (W5-8) can even watch the agents' own trades for errors.

3

Building & Governing It

The stack, the promise, the dangers

What It Takes to Build One

Five layers, from clean data up to human governance

What It Takes to Build One — the Stack

top

5 • Governance & oversight

human-in-the-loop, approval gates, audit logs, fairness & risk limits — stop "agent sprawl"

4 • Orchestration layer

the manager agent: assigns tasks, routes messages, decides who acts next

3 • The agents (LLM brains)

analyst, researcher, trader, risk — each an LLM with a role & memory (W12, W15)

2 • Tools the agents can call

search, code execution, backtester, the trading models & recommenders from this course

base

1 • Unified data platform

one clean source for prices, filings, positions — agents are only as good as the data they share

Firms that win aren't running fancier models — they're running them on a unified, governed platform.

The Boring Layer That Decides Everything

Agents are only as good as their shared data

Ask an agent to rebalance for ESG, and it must pull positions, check rules, run sims, and write reports.

If those live in disconnected systems, the agent gives poor answers. A unified data platform is the real moat.

Why it matters most

- Most pilots fail on plumbing, not on the model.
- Agents need one clean, machine-readable source.
- Good data > fancy models — Week 2's lesson, again.

The orchestration layer

- A manager agent assigns tasks and routes messages.
- It decides who acts next and when to stop.
- This is the hardest research problem in the field.

Real Promise, Real Dangers

Power and peril rise together

Real Promise, Real Dangers

The promise

Cover far more ground — every stock, all night, in parallel.

Consistent, repeatable process (W1).

Big cost savings on routine work.

A written rationale for every decision.

People shift from doing to deciding.

The dangers

Hallucination — confident, wrong (W12).

When agents can ACT, errors become trades.

Overfit backtests fool everyone (W10).

"Agent sprawl" — too many, ungoverned.

Herding: everyone's agents do the same thing.

Guardrails & humans are not optional.

When an agent can ACT, a mistake is no longer a wrong answer — it's a wrong trade.

Governance Is the Product

For money, trust is the whole game

The new failure modes

- Agent sprawl: dozens of unmanaged agents.
- Errors become actions (real trades, real losses).
- Herding: everyone's agents chase the same idea.
- Hidden bias and unfair outcomes at scale.

How serious firms respond

- A 'control tower': humans approve high-impact moves.
- Hard limits and kill-switches on what agents can do.
- Audit logs and a readable rationale for every action.
- Treat governance as infrastructure, built early.

High performers are far more likely to have defined human-in-the-loop checkpoints. That's the pattern.

Where Humans Stay in Charge

Autonomy by risk level, not all-or-nothing

1

Low-risk steps: run autonomously

Pulling data, drafting summaries, routine monitoring. Let the agents move fast where mistakes are cheap.

2

Medium-risk: human reviews

A proposed rebalance or a new thesis gets a human glance before it proceeds.

3

High-impact: human must approve

Large trades, client-facing actions, anything hard to undo. A person signs off, every time.

The goal isn't to remove humans — it's to spend their judgment where it matters most.



4

Your Turn

Sketch a firm, then simulate the team

This Firm Is the Whole Course, Assembled

Every week becomes one capability

Perception & decision

- RAG / LLM (W12): read filings, answer, explain.
- Models (W4-9): signals and anomaly detection.
- Recommenders (special): next best action for clients.

Action & control

- RL & execution (W13): trade without moving the price.
- Portfolios (W11): robust allocation (HRP/NCO).
- Honesty & fairness (W10, W14): the risk agent's job.

An AI-native firm isn't a new idea — it's your 16-week toolkit, wired into one cooperating system.

Don't Believe the Hype (Yet)

Powerful, early, and easy to get wrong

These systems are exciting and improving fast — but they are early, and markets are brutal teachers.

An agent that overfits or hallucinates can lose real money at machine speed. Discipline matters more, not less.

Stay grounded

- Backtests still lie (Week 10) — test out of sample.
- More autonomy = more ways to fool yourself.
- Start narrow; expand only what you can govern.

Where it genuinely helps today

- Research assistance and document-heavy work.
- Drafting, monitoring, and routine operations.
- Always with a human owning the final decision.

Getting Started

A tiny multi-agent firm, simulated in plain Python

A toy firm (no LLM needed)

- We mimic the agents as simple Python functions.
- Each 'agent' returns a view; the manager combines them.
- The point is the WORKFLOW, not real intelligence.
- Swap in real LLM calls later (the same shape).

What we'll do

- Lab 1: route a decision through analyst-risk-manager.
- Lab 2: add a risk VETO and watch it block a trade.
- All in a few lines of Python.
- See orchestration and human gates in miniature.

colab_setup.py — toy multi-agent firm (built inline)

```
# each agent is just a function returning a short 'view'
def analyst(stock): return {'signal': +1 if stock['news'] > 0 else -1}
def trader(signal): return {'order': 100 * signal}
def risk(order, limit=150): return abs(order) <= limit # True = ok
```

Hands-On Lab 1 — Route a Decision

Analyst -> Trader -> Risk -> Manager

Your task (step by step)

- 1. Analyst reads the 'news' and emits a signal.
- 2. Trader turns the signal into an order.
- 3. Risk checks it against a limit.
- 4. Manager approves only if risk says OK.
- Question: who has the final say?

`colab_lab1.py` • **toy multi-agent firm (built inline)**

```
stock = {'ticker': 'AAA', 'news': +0.5} # positive news

view = analyst(stock)           # 1 analyst
plan = trader(view['signal'])    # 2 trader
ok    = risk(plan['order'])      # 3 risk check

def manager(plan, ok):           # 4 orchestrator
    if ok: return f"APPROVED: trade {plan['order']}"
    return 'BLOCKED by risk'

print(manager(plan, ok))
```

Hands-On Lab 2 — Risk Veto & Human Gate

Big trades need a human; risky ones get blocked

Your task (step by step)

- 1. Make the trade large (over the limit).
- 2. Watch the risk agent veto it.
- 3. Add a human gate for big approved trades.
- 4. See autonomy scale with risk level.
- Question: which trades run on their own?

colab_lab2.py · toy multi-agent firm (built inline)

```
def govern(order):  
    if not risk(order):      # risk VETO  
        return 'BLOCKED: over risk limit'  
    if abs(order) > 50:      # high-impact -> human  
        return 'NEEDS HUMAN APPROVAL'  
    return 'AUTO-APPROVED'  # low-risk runs itself  
  
for order in [30, 120, 200]:  
    print(order, '->', govern(order))  
# 30 auto, 120 human, 200 blocked
```

Comprehension Check

Try first, then reveal the answer below

Q1 What best describes an 'AI-native' asset manager?

- A A firm with one big chatbot
- B** A firm whose workflow runs as a team of cooperating agents ✓
- C A firm with no computers
- D A faster spreadsheet

Answer: AI-native means the firm is built AROUND cooperating agents that run the workflow, with humans overseeing outcomes.

Q2 Why is governance so critical for agentic investing?

- A It makes agents faster
- B** Because when agents can act, errors become real trades ✓
- C It removes all humans
- D It lowers data quality

Answer: An acting agent can execute mistakes at machine speed, so guardrails, limits, and human approval are essential.

Three Ideas, One Picture Each

1**AI-native = built around agents**

Not AI as a tool you prompt, but a team of agents that run the workflow while humans oversee outcomes.

2**A specialist team, with debate**

Analyst, researcher, trader, risk, and a manager-orchestrator — mirroring a real firm, leaving an audit trail.

3**Governance is the product**

Unified data, human-in-the-loop gates by risk level, hard limits, and audit logs. For money, trust is everything.

It's the entire course wired into one system — perception, decision, action, and control.

Special Session — Three Things to Remember

1

The shift is from tool to teammate

AI-native firms rebuild the workflow around cooperating agents. People move from doing tasks to deciding.

2

Structure beats a single mind

Specialist roles that hand off and debate, separating decide from execute, give sharper, checkable decisions.

3

Build on data, win on governance

A unified data platform is the moat; human gates, limits, and audit trails make it safe enough for real money.

More autonomy raises the stakes, not the rules. Discipline from Weeks 1, 10, and 14 matters more than ever.

Try both labs, then sketch your own agent firm: list the roles, their tools, and the human gates.